



3x

FASTER THAN QUICKFIX
246 ns vs 730 ns mean parse

7.5x

FASTER THAN FIX8
193 ns vs 1,453 ns P50 latency

4.17M

MESSAGES / SECOND
687 MB/s bandwidth

0

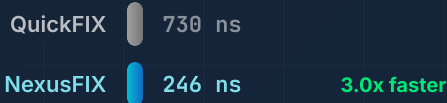
HEAP ALLOCATIONS
Zero-alloc critical path

CORE ADVANTAGES

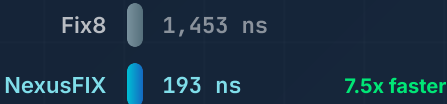
- ✓ **Zero-Copy Parsing** — std::span views into raw buffer, no string copies
- ✓ **O(1) Field Lookup** — Direct array indexing by FIX tag number
- ✓ **SIMD Scanning** — AVX2 delimiter detection, up to 14.7x on large buffers
- ✓ **Compile-Time Offsets** — consteval field

PERFORMANCE BENCHMARKS

EXECUTIONREPORT PARSE LATENCY (MEAN)



P50 LATENCY VS FIX8



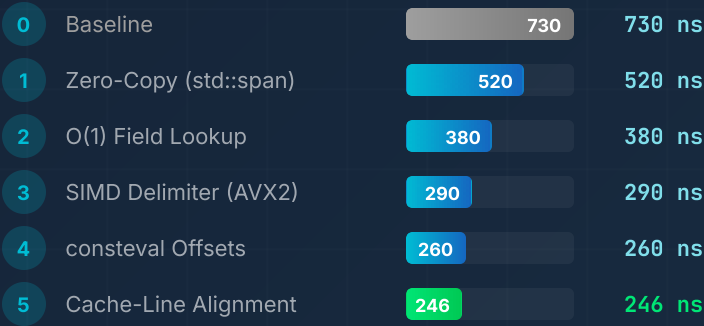
THROUGHPUT (MSG/SEC)



P99 TAIL LATENCY



OPTIMIZATION JOURNEY: 730 → 246 NS



API EXAMPLE

```
// Zero-copy FIX message parsing
FixParser parser;
auto result = parser.parse(raw_buffer);
if (result) {
    auto id = result→get_string(Tag::OrderID);
    auto qty = result→get_qty(Tag::LastQty);
    // O(1) lookup, zero heap alloc
}
```

tables, zero
runtime cost

✓ **No Heap
Allocation** —
PMR arena pools
+ SessionHeap
(mimalloc)

✓ **Cache-Line
Aligned** —
alignas(64),
branch hints,
LTO

♦ MODERN C++23

constexpr

std::span

std::expected

concepts

std::variant

string_view

PMR

[[likely]]

assume_aligned

co_await

```
}
```

```
// Builder pattern for orders
auto msg = NewOrderSingle::Builder{}
    .symbol("AAPL")
    .side(Side::Buy)
    .order_qty(100_qty)
    .price(150.50_price)
    .build(asm_);
```

▣ ARCHITECTURE

★ USE CASES

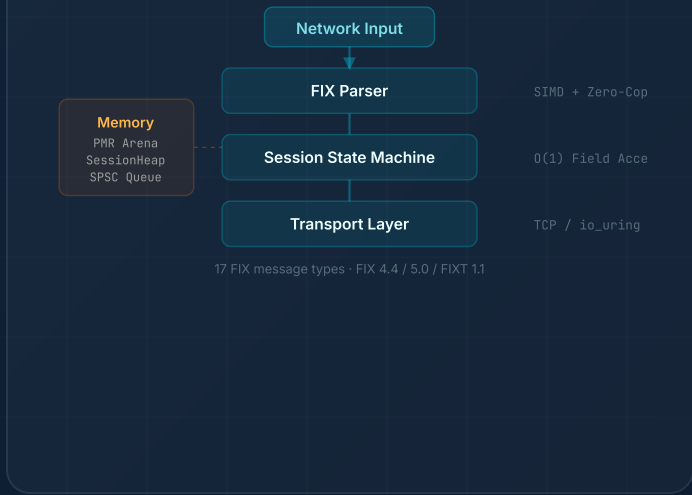
⚡ **High-Frequency Trading**
Sub-microsecond order execution

⚖ **Low-Latency Strategy Engines**
Deterministic, jitter-free execution

↗ RESOURCES & LINKS



GitHub Repository
[github.com/SilverstreamsA
I/NexusFix](https://github.com/SilverstreamsAI/NexusFix)



Market Data Pipelines
FIX-based feed processing at 4M+ msg/sec



FIX Protocol Education
Modern C++23 reference implementation



Benchmark Comparison
github.com/.../docs/comparison



Optimization Diary
730ns to 246ns – 5-phase journey



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NexusFIX — Built with Modern C++23 · GCC 13+ / Clang 17+ · CMake 3.20+ · Linux / macOS / Windows

#HFT

#FIX

#C++23

#ZeroAlloc

#SIMD

#LowLatency